

# 1c\_behavior

Auto-generated from 2\_pipeline/1c\_behavior/out/ on 2026-09-08 10:42. Re-run make\_progress\_md.py after the script's output changes -- don't hand-edit this file.

## Tables

### Behavior mi vs linear

| protocol | window   | var_x      | var_y        | n_trials | r        | r2        | p_linear   |
|----------|----------|------------|--------------|----------|----------|-----------|------------|
| DM       | pre_stim | zpos       | runspeed     | 3045     | 0.3281   | 0.1077    | 2.382e-05  |
| DM       | pre_stim | zpos       | lick         | 3045     | -0.2123  | 0.04507   | 2.263e-05  |
| DM       | pre_stim | zpos       | pupil_area   | 988      | 0.1159   | 0.01344   | 0.00026    |
| DM       | pre_stim | zpos       | pupil_xpos   | 988      | -0.05994 | 0.003593  | 0.05966    |
| DM       | pre_stim | zpos       | pupil_ypos   | 988      | -0.1168  | 0.01363   | 0.00023    |
| DM       | pre_stim | zpos       | motionenergy | 2014     | -0.1064  | 0.01131   | 1.723e-05  |
| DM       | pre_stim | zpos       | videoPC_0    | 2014     | -0.08288 | 0.006868  | 0.00019    |
| DM       | pre_stim | zpos       | videoPC_1    | 2014     | 0.0595   | 0.00354   | 0.00756    |
| DM       | pre_stim | runspeed   | lick         | 3045     | -0.3765  | 0.1418    | 3.738e-103 |
| DM       | pre_stim | runspeed   | pupil_area   | 988      | 0.1297   | 0.01683   | 4.323e-05  |
| DM       | pre_stim | runspeed   | pupil_xpos   | 988      | -0.04729 | 0.002236  | 0.1375     |
| DM       | pre_stim | runspeed   | pupil_ypos   | 988      | -0.0166  | 0.0002757 | 0.6022     |
| DM       | pre_stim | runspeed   | motionenergy | 2014     | -0.06886 | 0.004742  | 0.00198    |
| DM       | pre_stim | runspeed   | videoPC_0    | 2014     | 0.13     | 0.0169    | 4.741e-05  |
| DM       | pre_stim | runspeed   | videoPC_1    | 2014     | 0.07381  | 0.005447  | 0.00091    |
| DM       | pre_stim | lick       | pupil_area   | 988      | -0.03362 | 0.00113   | 0.2911     |
| DM       | pre_stim | lick       | pupil_xpos   | 988      | 0.01964  | 0.0003859 | 0.5374     |
| DM       | pre_stim | lick       | pupil_ypos   | 988      | -0.01686 | 0.0002843 | 0.5966     |
| DM       | pre_stim | lick       | motionenergy | 2014     | 0.1503   | 0.0226    | 1.2e-11    |
| DM       | pre_stim | lick       | videoPC_0    | 2014     | 0.0813   | 0.00661   | 0.00025    |
| DM       | pre_stim | lick       | videoPC_1    | 2014     | 0.1439   | 0.0207    | 8.748e-05  |
| DM       | pre_stim | pupil_area | pupil_xpos   | 988      | 0.3471   | 0.1205    | 2.383e-05  |

| protocol | window   | var_x      | var_y        | n_trials | r       | r2        | p_linea    |
|----------|----------|------------|--------------|----------|---------|-----------|------------|
| DM       | pre_stim | pupil_area | pupil_ypos   | 988      | -0.3645 | 0.1329    | 2.08e-3    |
| DM       | pre_stim | pupil_area | motionenergy | 988      | 0.111   | 0.01232   | 0.00047    |
| DM       | pre_stim | pupil_area | videoPC_0    | 988      | -0.2302 | 0.05301   | 2.366e-    |
| DM       | pre_stim | pupil_area | videoPC_1    | 988      | 0.03088 | 0.0009533 | 0.3323     |
| DM       | pre_stim | pupil_xpos | pupil_ypos   | 988      | -0.7729 | 0.5973    | 5.681e-197 |
| DM       | pre_stim | pupil_xpos | motionenergy | 988      | 0.3244  | 0.1053    | 1.195e-    |
| DM       | pre_stim | pupil_xpos | videoPC_0    | 988      | -0.4899 | 0.24      | 9.139e-    |
| DM       | pre_stim | pupil_xpos | videoPC_1    | 988      | -0.1441 | 0.02077   | 5.42e-0    |

(showing first 15 of 16 columns; showing first 30 of 324 rows -- see the linked CSV for the rest)

[behavior\\_mi\\_vs\\_linear.csv](#)

## Behavior multivariate mi vs linear

| protocol | window   | target   | predictors                                                                                                  | n_predictors | n_trials | r2     | adj |
|----------|----------|----------|-------------------------------------------------------------------------------------------------------------|--------------|----------|--------|-----|
| DM       | pre_stim | zpos     | runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1 | 8            | 988      | 0.2451 | 0.2 |
| DM       | pre_stim | runspeed | zpos, lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1        | 8            | 988      | 0.4273 | 0.4 |
| DM       | pre_stim | lick     | zpos,<br>runspeed,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,                            | 8            | 988      | 0.3173 | 0.3 |

| protocol | window   | target       | predictors                                                                                             | n_predictors | n_trials | r2     | adj |
|----------|----------|--------------|--------------------------------------------------------------------------------------------------------|--------------|----------|--------|-----|
|          |          |              | videoPC_0,<br>videoPC_1                                                                                |              |          |        |     |
| DM       | pre_stim | pupil_area   | zpos,<br>runspeed,<br>lick,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1  | 8            | 988      | 0.1934 | 0.1 |
| DM       | pre_stim | pupil_xpos   | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1  | 8            | 988      | 0.6498 | 0.6 |
| DM       | pre_stim | pupil_ypos   | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1  | 8            | 988      | 0.6693 | 0.6 |
| DM       | pre_stim | motionenergy | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>videoPC_0,<br>videoPC_1    | 8            | 988      | 0.3526 | 0.3 |
| DM       | pre_stim | videoPC_0    | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_1 | 8            | 988      | 0.527  | 0.5 |

| protocol | window   | target     | predictors                                                                                                  | n_predictors | n_trials | r2     | adj |
|----------|----------|------------|-------------------------------------------------------------------------------------------------------------|--------------|----------|--------|-----|
| DM       | pre_stim | videoPC_1  | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0      | 8            | 988      | 0.2677 | 0.2 |
| DM       | stim     | zpos       | runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1 | 8            | 990      | 0.2659 | 0.2 |
| DM       | stim     | runspeed   | zpos, lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1        | 8            | 990      | 0.5507 | 0.5 |
| DM       | stim     | lick       | zpos,<br>runspeed,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1 | 8            | 990      | 0.536  | 0.5 |
| DM       | stim     | pupil_area | zpos,<br>runspeed,<br>lick,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1       | 8            | 990      | 0.2637 | 0.2 |
| DM       | stim     | pupil_xpos | zpos,<br>runspeed,<br>lick,<br>pupil_area,                                                                  | 8            | 990      | 0.6747 | 0.6 |



| protocol | window | target       | predictors                                                                                             | n_predictors | n_trials | r2     | adj |
|----------|--------|--------------|--------------------------------------------------------------------------------------------------------|--------------|----------|--------|-----|
|          |        |              | pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1                                                |              |          |        |     |
| DM       | stim   | pupil_ypos   | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1  | 8            | 990      | 0.6878 | 0.6 |
| DM       | stim   | motionenergy | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>videoPC_0,<br>videoPC_1    | 8            | 990      | 0.355  | 0.3 |
| DM       | stim   | videoPC_0    | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_1 | 8            | 990      | 0.197  | 0.1 |
| DM       | stim   | videoPC_1    | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0 | 8            | 990      | 0.241  | 0.2 |
| DM       | reward | zpos         | runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,                       | 8            | 989      | 0.1452 | 0.1 |

| protocol | window | target     | predictors                                                                                                  | n_predictors | n_trials | r2     | adj |
|----------|--------|------------|-------------------------------------------------------------------------------------------------------------|--------------|----------|--------|-----|
|          |        |            | videoPC_0,<br>videoPC_1                                                                                     |              |          |        |     |
| DM       | reward | runspeed   | zpos, lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1        | 8            | 989      | 0.5168 | 0.5 |
| DM       | reward | lick       | zpos,<br>runspeed,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1 | 8            | 989      | 0.5674 | 0.5 |
| DM       | reward | pupil_area | zpos,<br>runspeed,<br>lick,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1       | 8            | 989      | 0.2513 | 0.2 |
| DM       | reward | pupil_xpos | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1       | 8            | 989      | 0.606  | 0.6 |
| DM       | reward | pupil_ypos | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1       | 8            | 989      | 0.6273 | 0.6 |

| protocol | window   | target       | predictors                                                                                                  | n_predictors | n_trials | r2     | adj |
|----------|----------|--------------|-------------------------------------------------------------------------------------------------------------|--------------|----------|--------|-----|
| DM       | reward   | motionenergy | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>videoPC_0,<br>videoPC_1         | 8            | 989      | 0.5771 | 0.5 |
| DM       | reward   | videoPC_0    | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_1      | 8            | 989      | 0.4078 | 0.4 |
| DM       | reward   | videoPC_1    | zpos,<br>runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0      | 8            | 989      | 0.1667 | 0.1 |
| DP       | pre_stim | zpos         | runspeed,<br>lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1 | 8            | 1137     | 0.183  | 0.1 |
| DP       | pre_stim | runspeed     | zpos, lick,<br>pupil_area,<br>pupil_xpos,<br>pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1        | 8            | 1137     | 0.5521 | 0.5 |
| DP       | pre_stim | lick         | zpos,<br>runspeed,<br>pupil_area,<br>pupil_xpos,                                                            | 8            | 1137     | 0.2985 | 0.2 |

| protocol | window | target | predictors                                              | n_predictors | n_trials | r2 | adj |
|----------|--------|--------|---------------------------------------------------------|--------------|----------|----|-----|
|          |        |        | pupil_ypos,<br>motionenergy,<br>videoPC_0,<br>videoPC_1 |              |          |    |     |

(showing first 15 of 17 columns; showing first 30 of 81 rows -- see the linked CSV for the rest)

[behavior\\_multivariate\\_mi\\_vs\\_linear.csv](#)

## Behavior multivariate stepwise

| protocol | window   | target   | step | predictor_added | cumulative_r2 | cumulative_Δ |
|----------|----------|----------|------|-----------------|---------------|--------------|
| DM       | pre_stim | zpos     | 1    | runspeed        | 0.1284        | 0.1276       |
| DM       | pre_stim | zpos     | 2    | videoPC_0       | 0.1686        | 0.167        |
| DM       | pre_stim | zpos     | 3    | pupil_xpos      | 0.1949        | 0.1925       |
| DM       | pre_stim | zpos     | 4    | pupil_ypos      | 0.2257        | 0.2225       |
| DM       | pre_stim | zpos     | 5    | motionenergy    | 0.2377        | 0.2339       |
| DM       | pre_stim | zpos     | 6    | videoPC_1       | 0.2433        | 0.2386       |
| DM       | pre_stim | zpos     | 7    | pupil_area      | 0.2451        | 0.2397       |
| DM       | pre_stim | zpos     | 8    | lick            | 0.2451        | 0.2389       |
| DM       | pre_stim | runspeed | 1    | lick            | 0.1835        | 0.1827       |
| DM       | pre_stim | runspeed | 2    | zpos            | 0.2564        | 0.2549       |
| DM       | pre_stim | runspeed | 3    | videoPC_0       | 0.3258        | 0.3238       |
| DM       | pre_stim | runspeed | 4    | motionenergy    | 0.4033        | 0.4009       |
| DM       | pre_stim | runspeed | 5    | pupil_area      | 0.4236        | 0.4206       |
| DM       | pre_stim | runspeed | 6    | pupil_xpos      | 0.4258        | 0.4223       |
| DM       | pre_stim | runspeed | 7    | videoPC_1       | 0.4273        | 0.4232       |
| DM       | pre_stim | runspeed | 8    | pupil_ypos      | 0.4273        | 0.4226       |
| DM       | pre_stim | lick     | 1    | runspeed        | 0.1835        | 0.1827       |
| DM       | pre_stim | lick     | 2    | videoPC_1       | 0.2253        | 0.2238       |
| DM       | pre_stim | lick     | 3    | motionenergy    | 0.2714        | 0.2692       |
| DM       | pre_stim | lick     | 4    | videoPC_0       | 0.3132        | 0.3104       |
| DM       | pre_stim | lick     | 5    | pupil_area      | 0.3158        | 0.3123       |

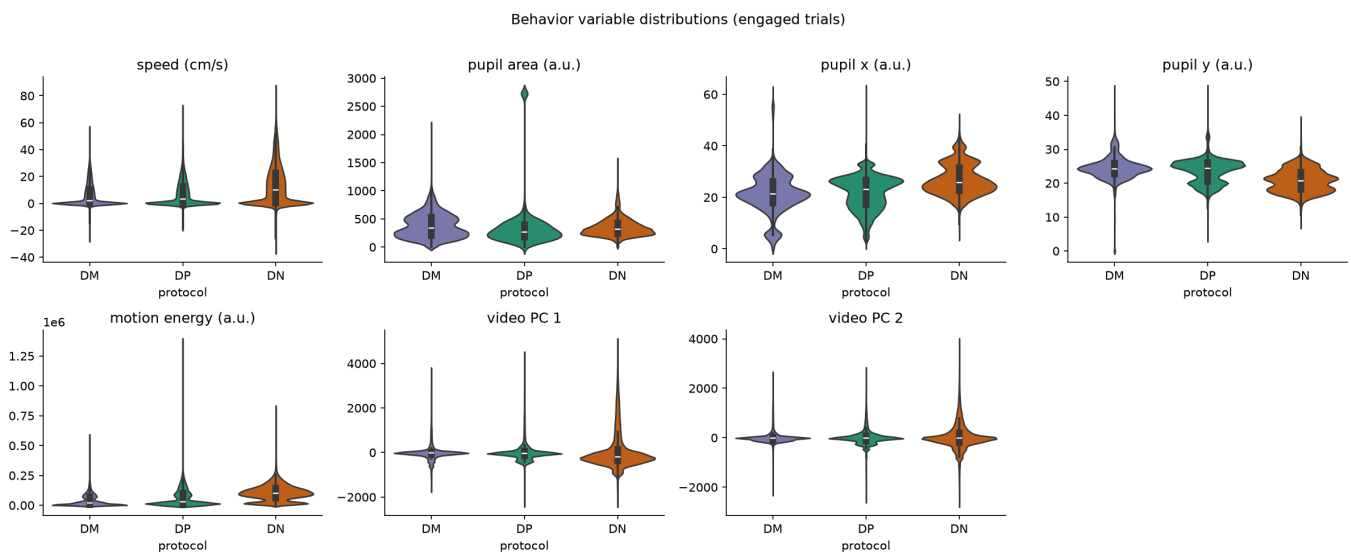
| protocol | window   | target     | step | predictor_added | cumulative_r2 | cumulative_ε |
|----------|----------|------------|------|-----------------|---------------|--------------|
| DM       | pre_stim | lick       | 6    | pupil_xpos      | 0.3172        | 0.3131       |
| DM       | pre_stim | lick       | 7    | zpos            | 0.3173        | 0.3124       |
| DM       | pre_stim | lick       | 8    | pupil_ypos      | 0.3173        | 0.3117       |
| DM       | pre_stim | pupil_area | 1    | pupil_ypos      | 0.1329        | 0.132        |
| DM       | pre_stim | pupil_area | 2    | runspeed        | 0.1482        | 0.1464       |
| DM       | pre_stim | pupil_area | 3    | pupil_xpos      | 0.1614        | 0.1589       |
| DM       | pre_stim | pupil_area | 4    | videoPC_1       | 0.17          | 0.1666       |
| DM       | pre_stim | pupil_area | 5    | videoPC_0       | 0.1846        | 0.1805       |
| DM       | pre_stim | pupil_area | 6    | motionenergy    | 0.1897        | 0.1848       |

(showing first 30 of 648 rows -- see the linked CSV for the rest)

[behavior\\_multivariate\\_stepwise.csv](#)

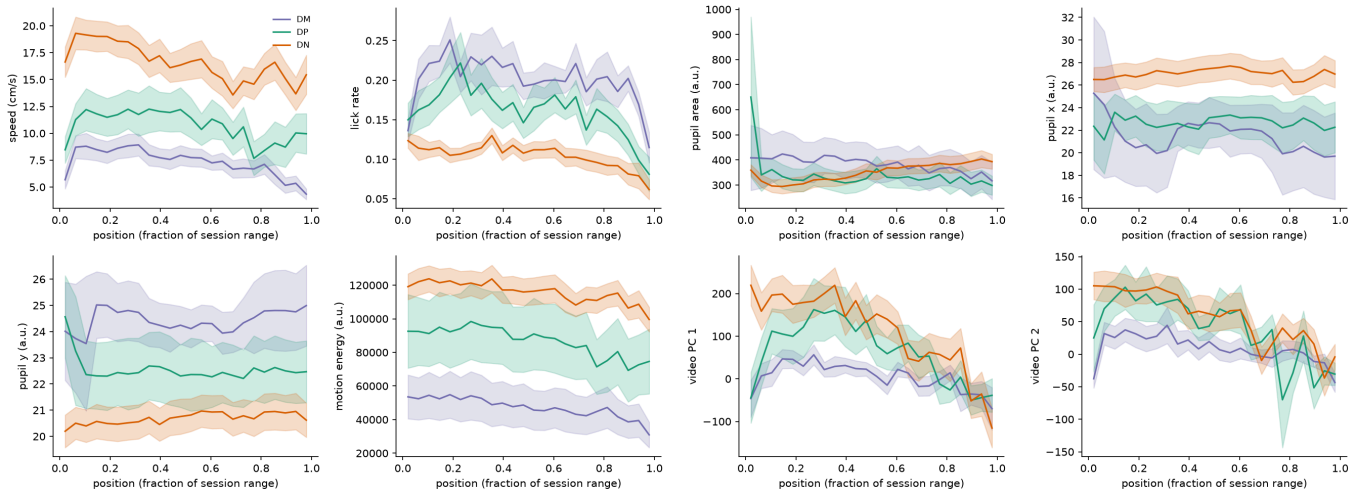
## Figures

### Variable distributions



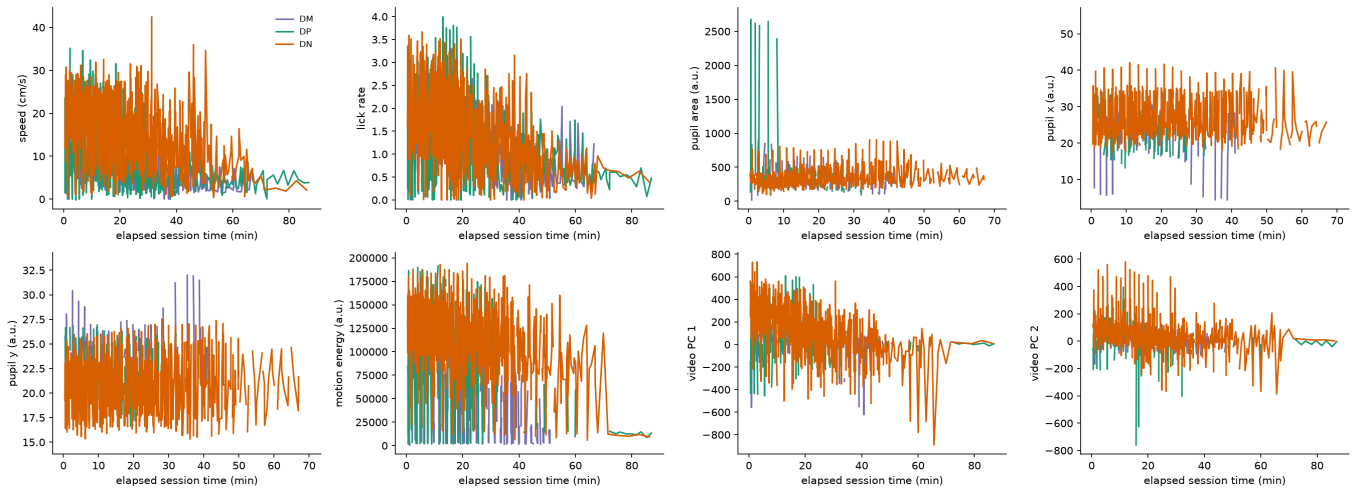
### Corridor position tuning

Behavior variables vs. position, whole session (engaged trials)

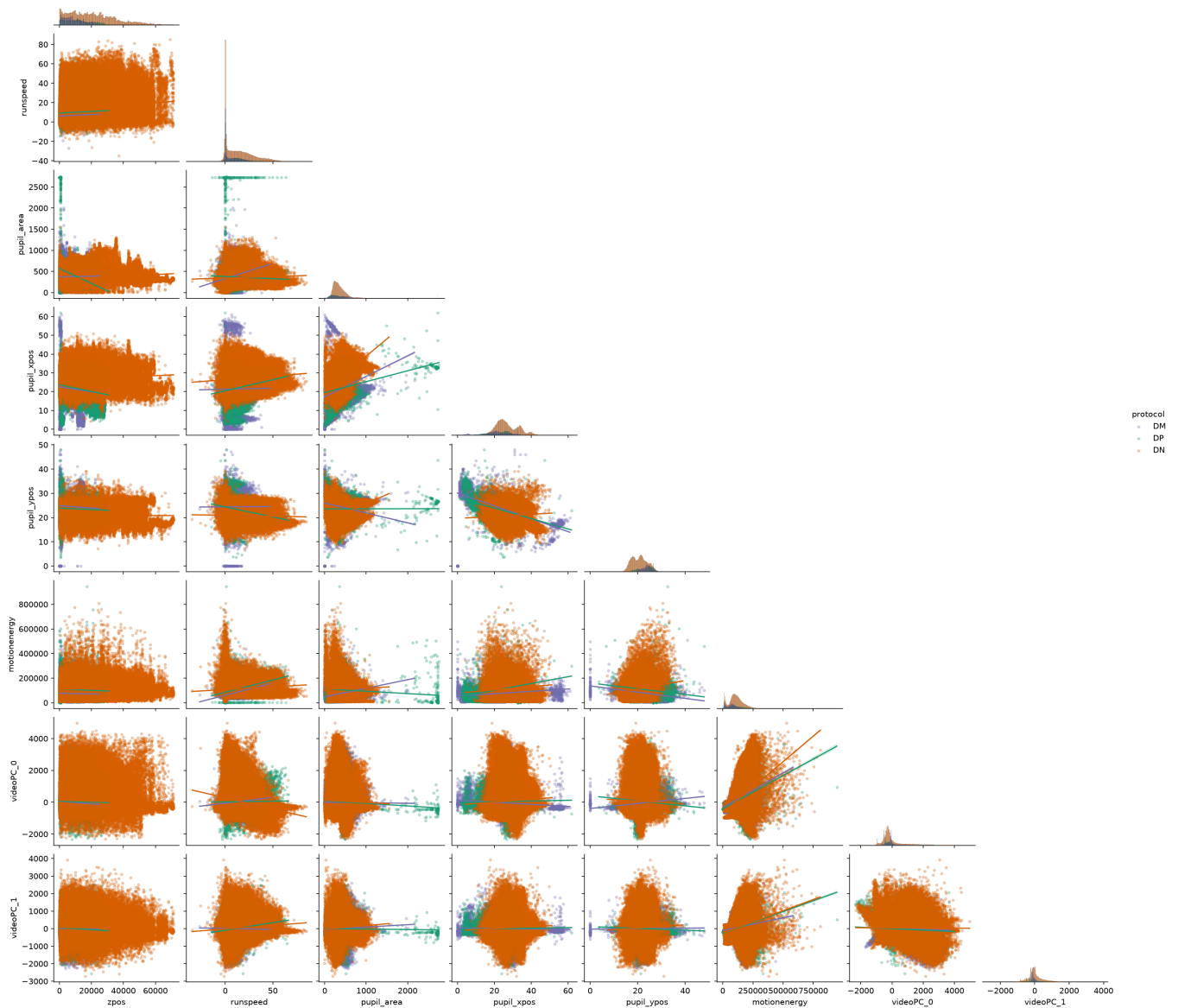


## 2b session time course

Behavior variables vs. elapsed session time (engaged trials)

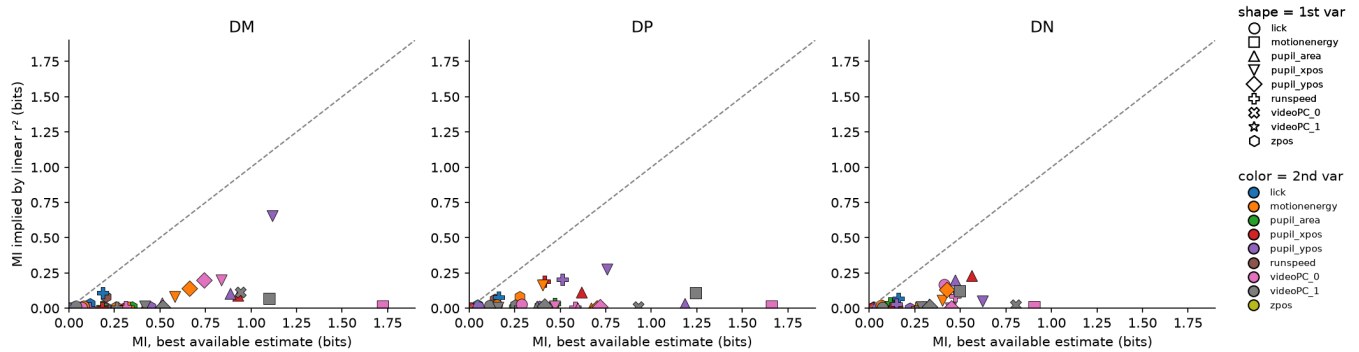


## Pairwise relationships

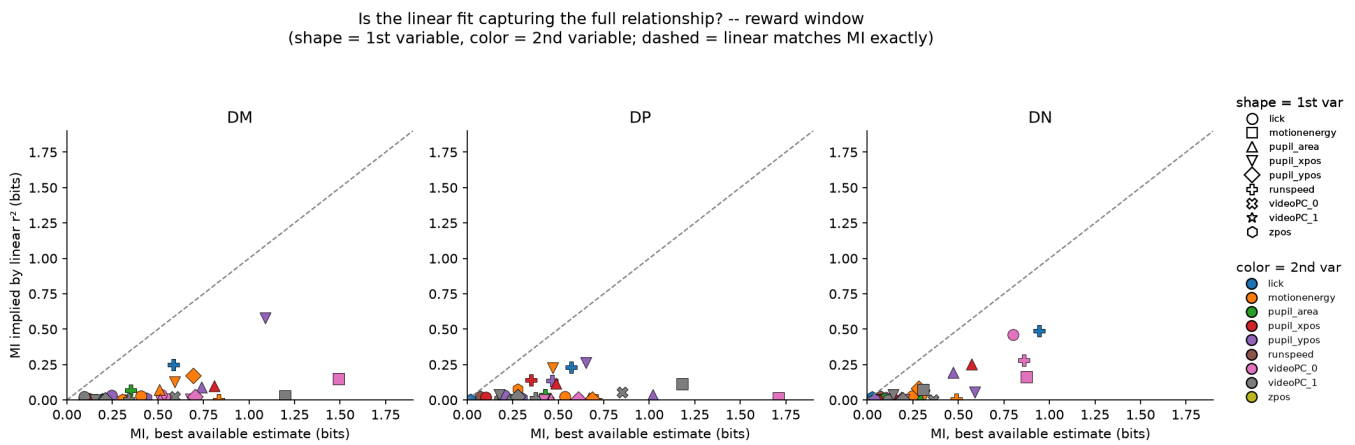


## Mi vs linear summary pre stim

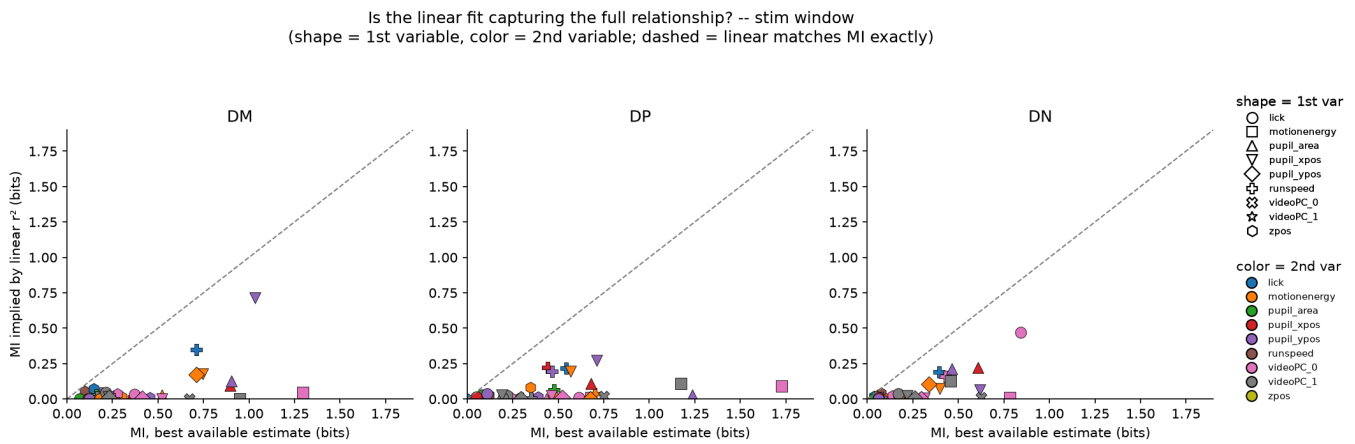
Is the linear fit capturing the full relationship? -- pre\_stim window  
(shape = 1st variable, color = 2nd variable; dashed = linear matches MI exactly)



# Mi vs linear summary reward



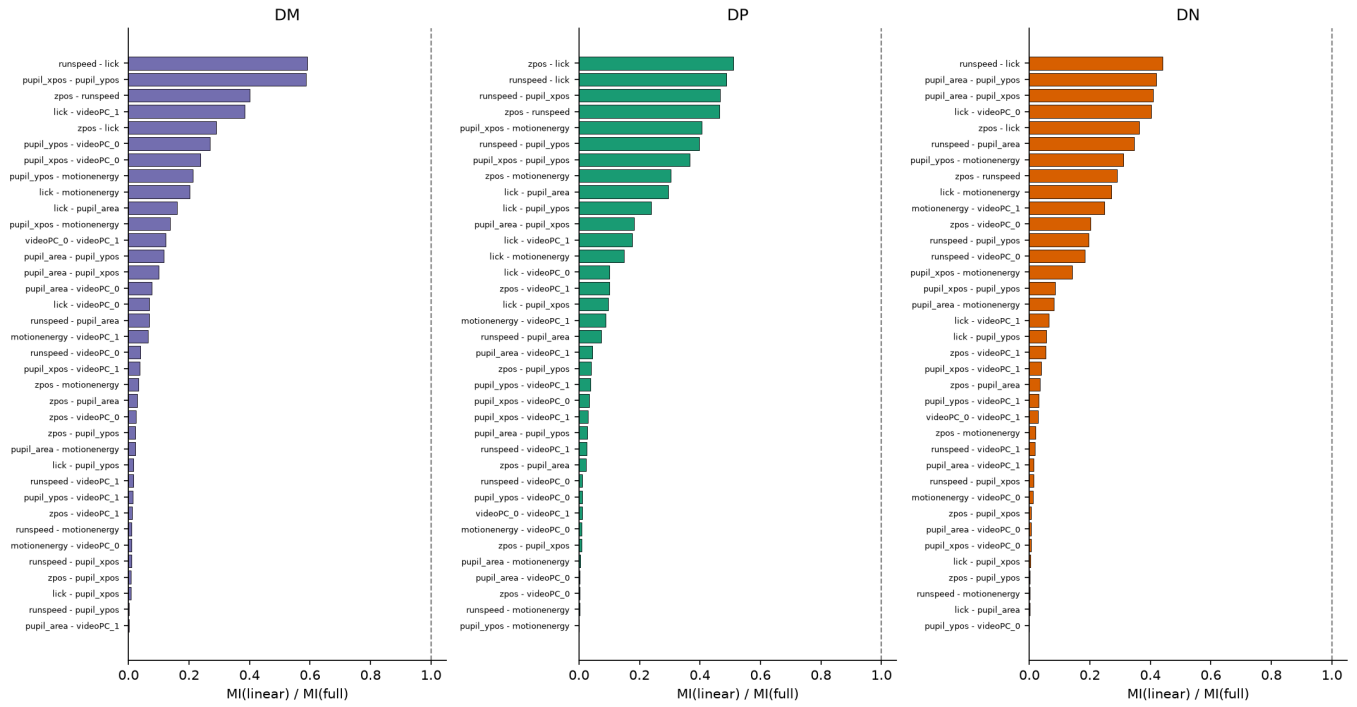
# Mi vs linear summary stim



# 4a frac explained linearly pre stim

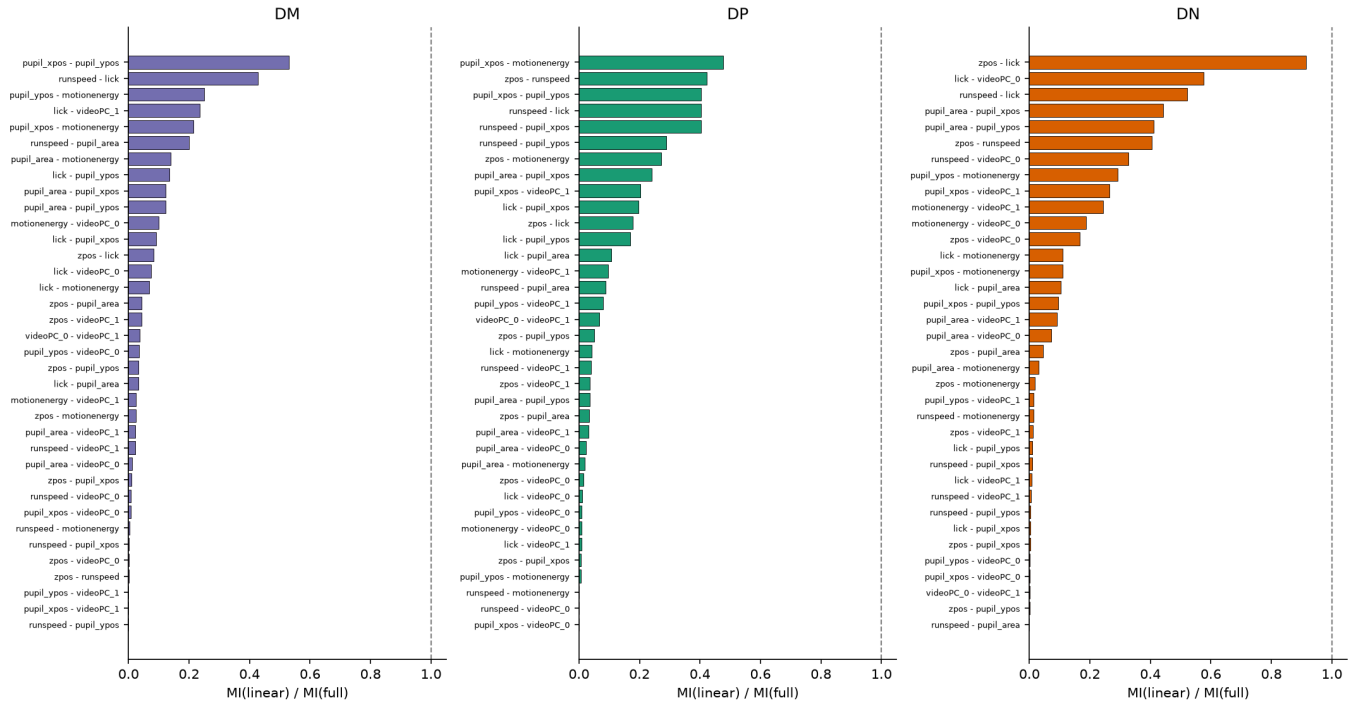


Fraction of mutual information a linear fit captures, per pair -- pre\_stim window



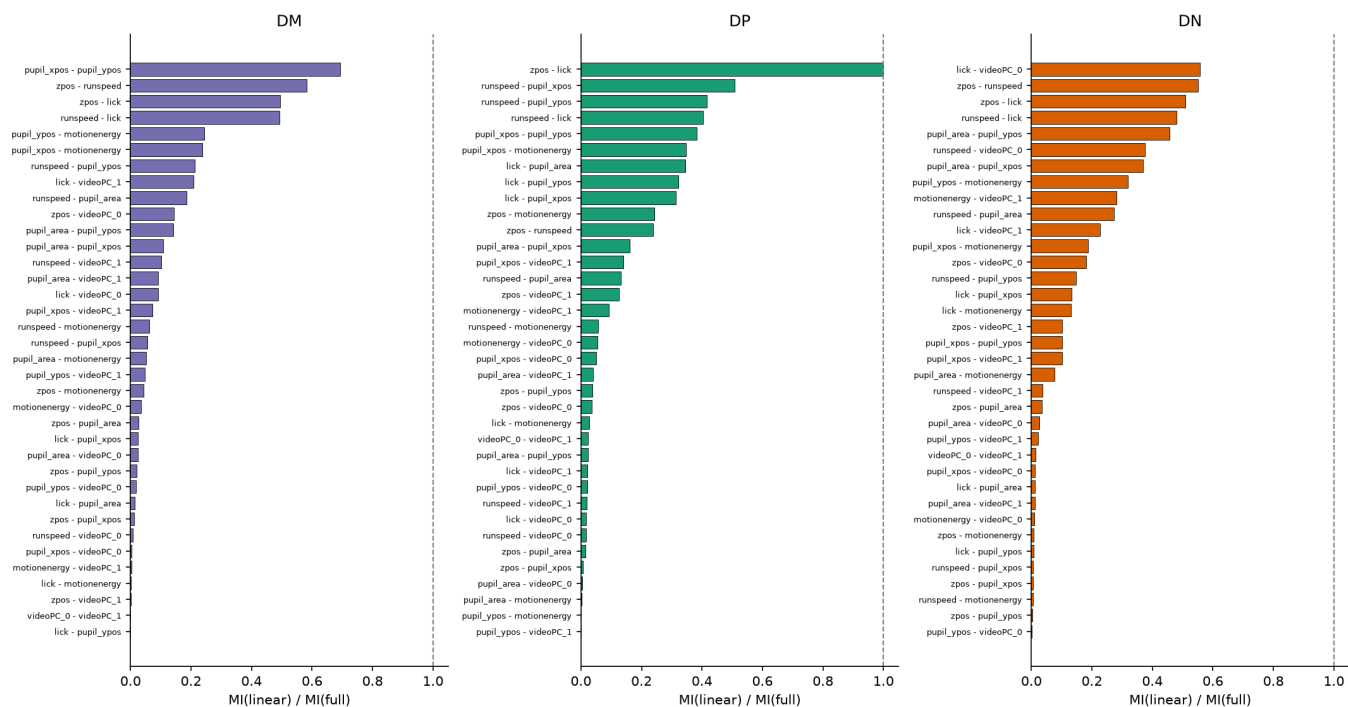
## 4a frac explained linearly reward

Fraction of mutual information a linear fit captures, per pair -- reward window



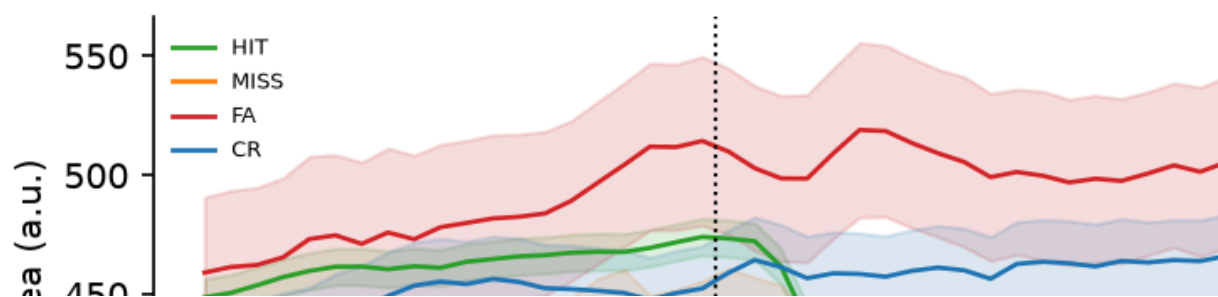
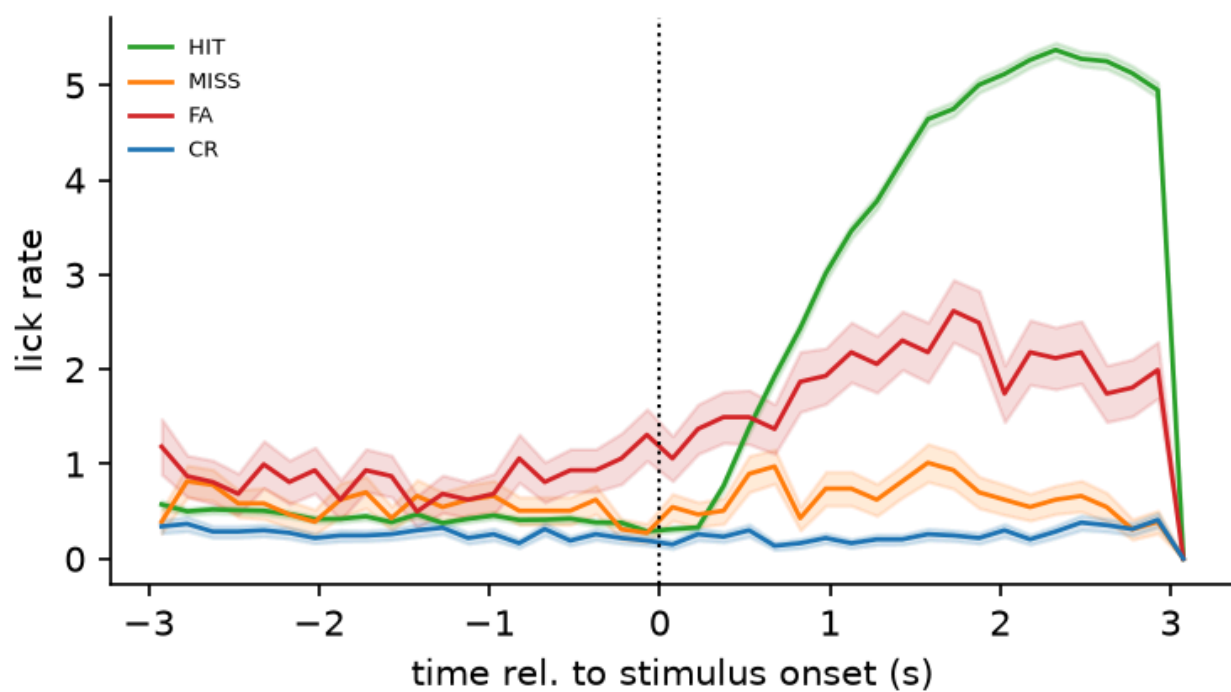
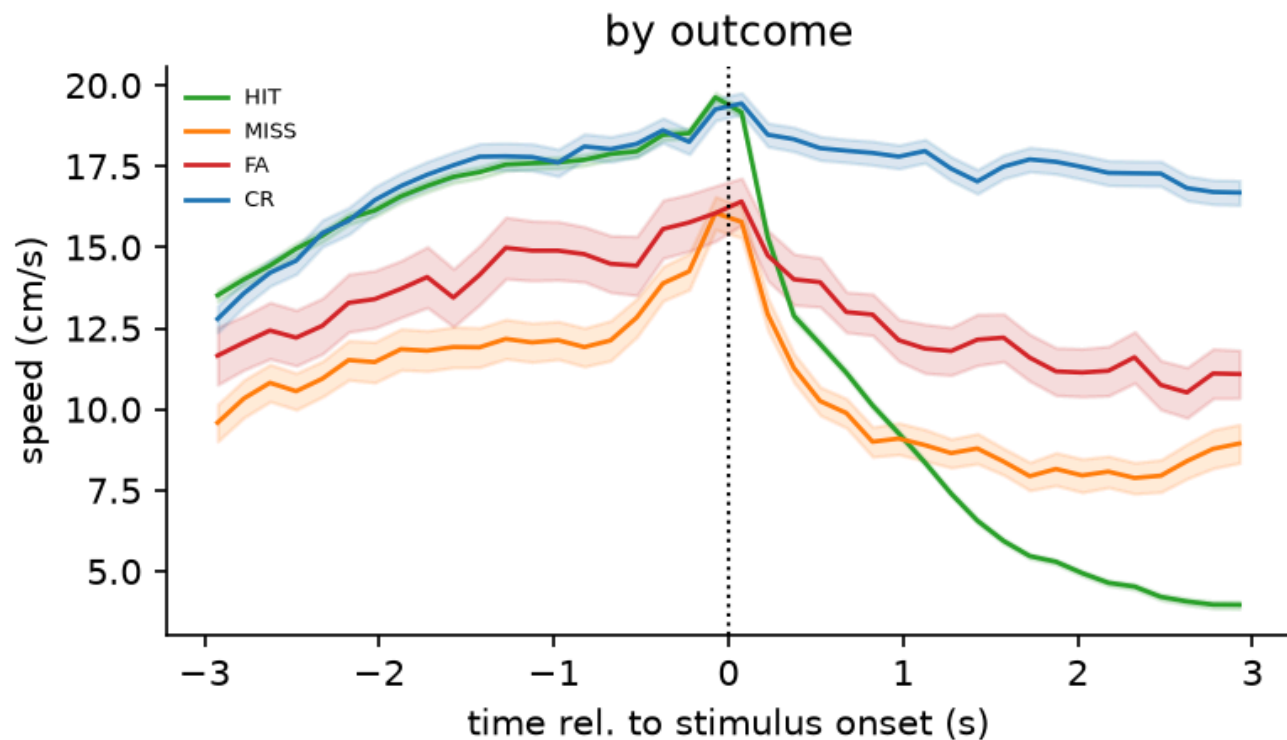
## 4a frac explained linearly stim

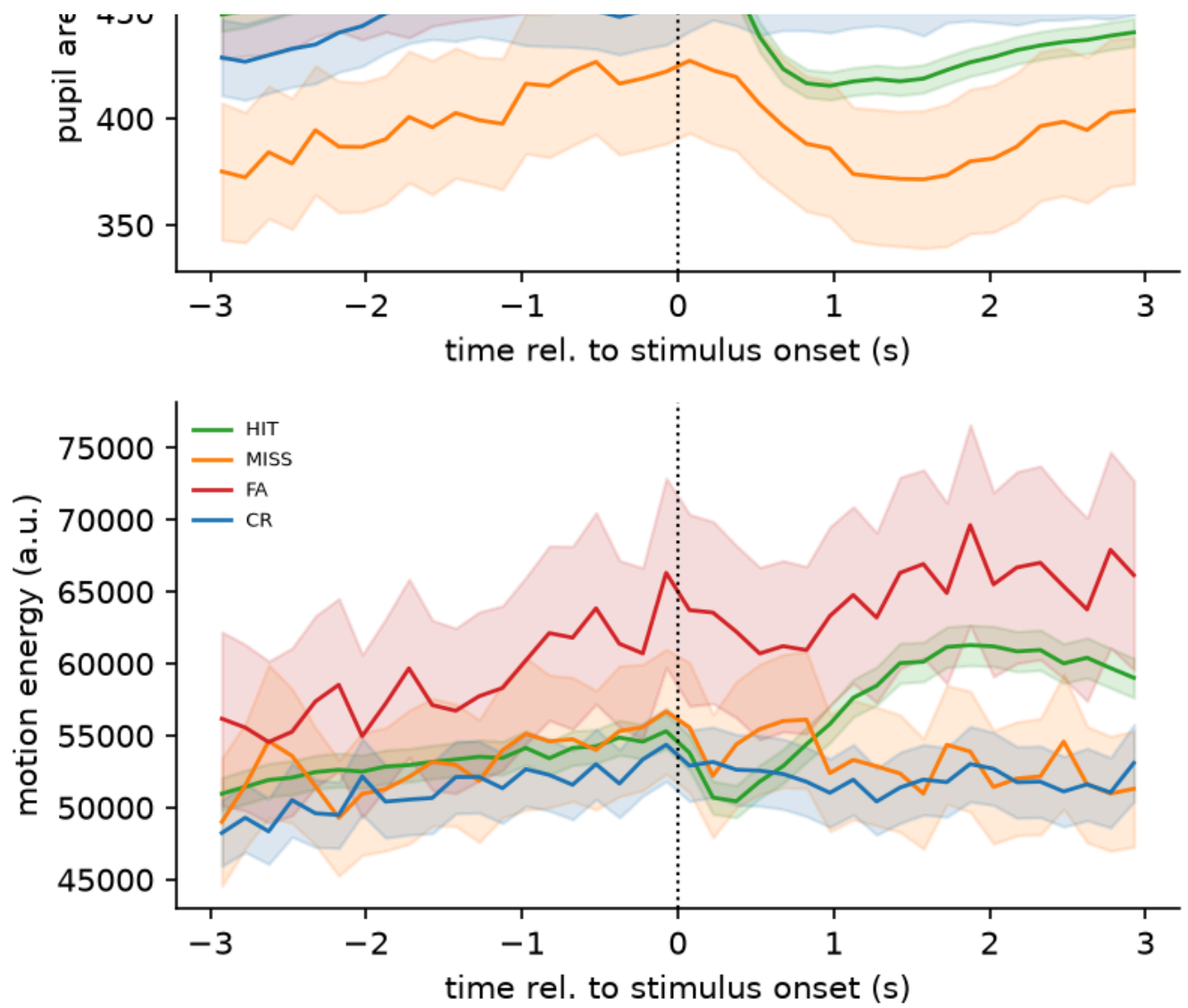
Fraction of mutual information a linear fit captures, per pair -- stim window



Psth time DM

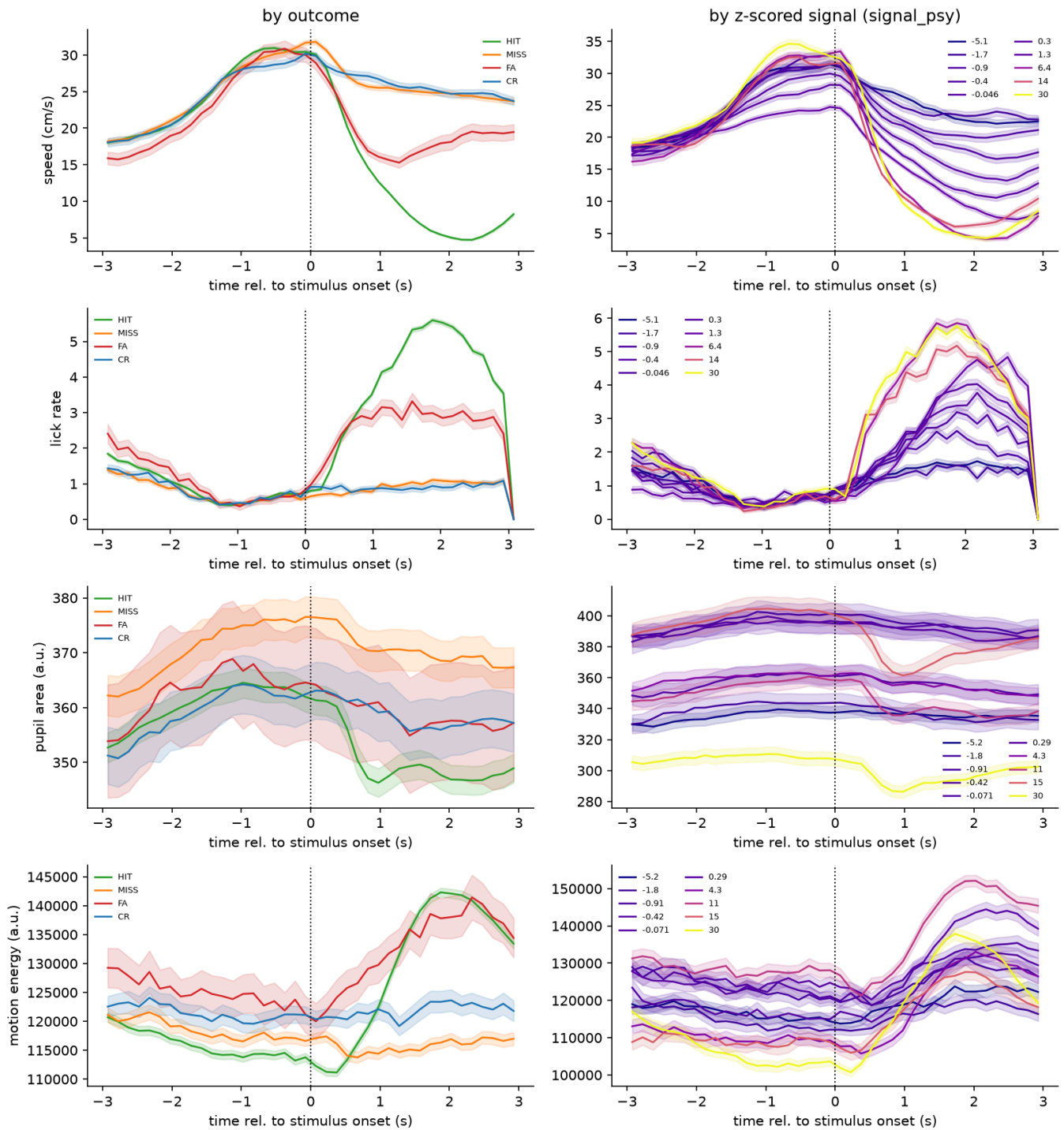
Trial-aligned time -- DM (0 = stimulus onset)





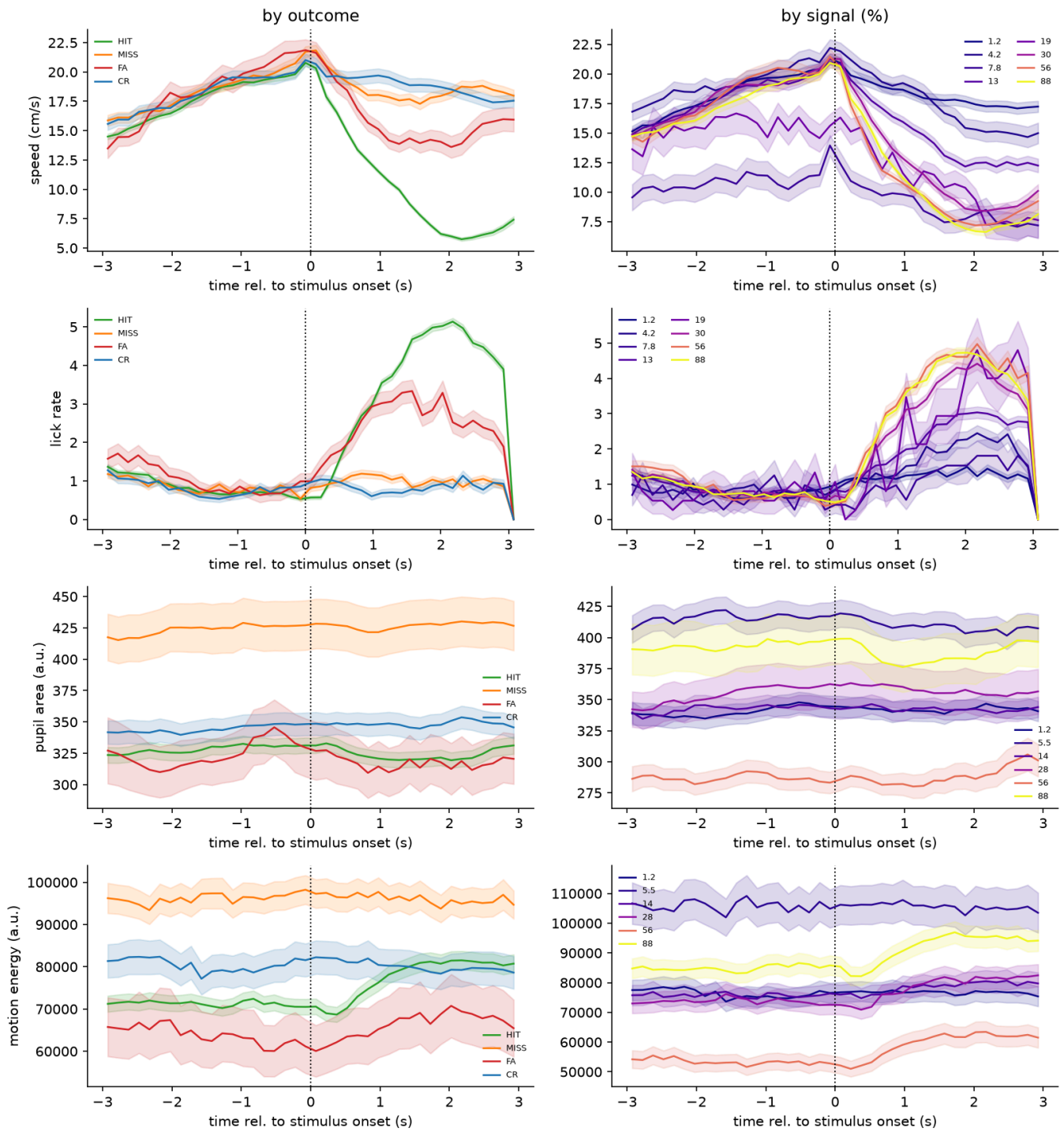
Psth time DN

Trial-aligned time -- DN (0 = stimulus onset)



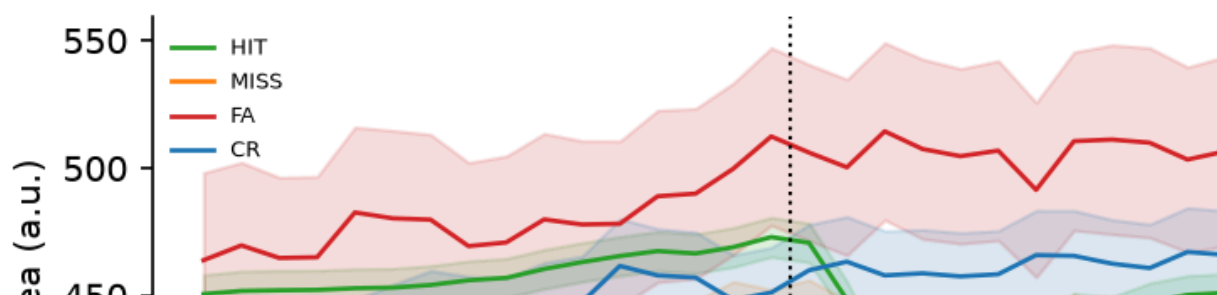
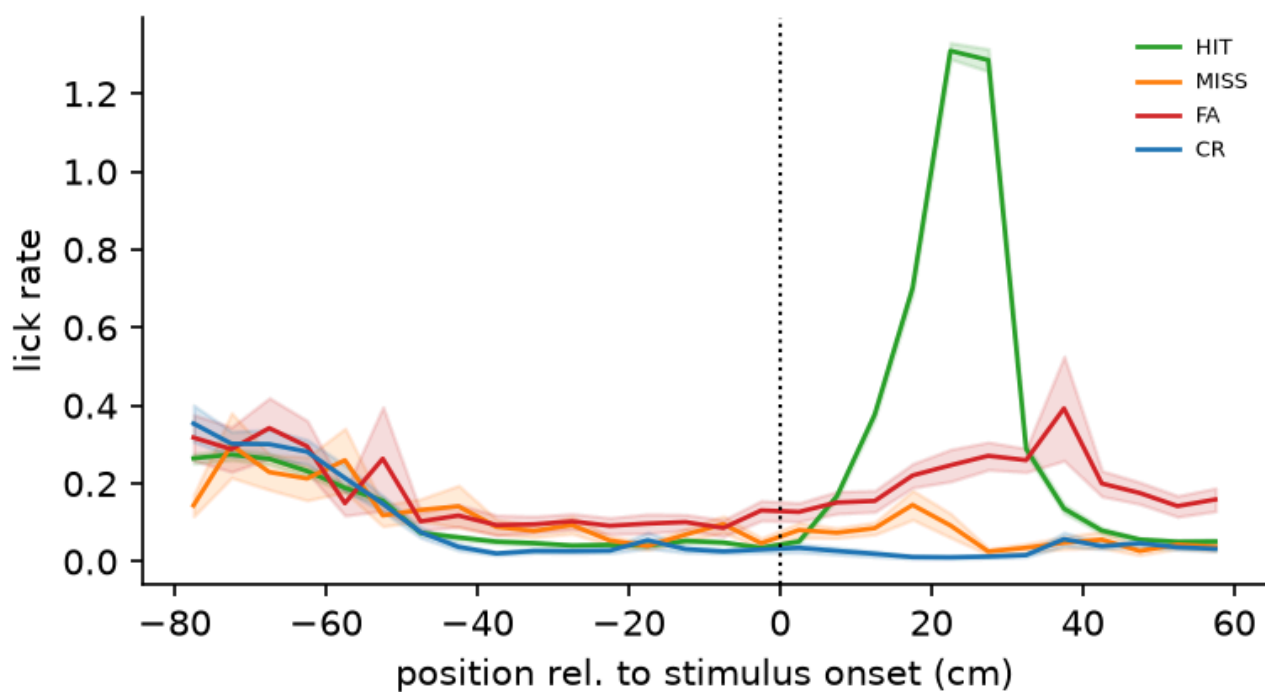
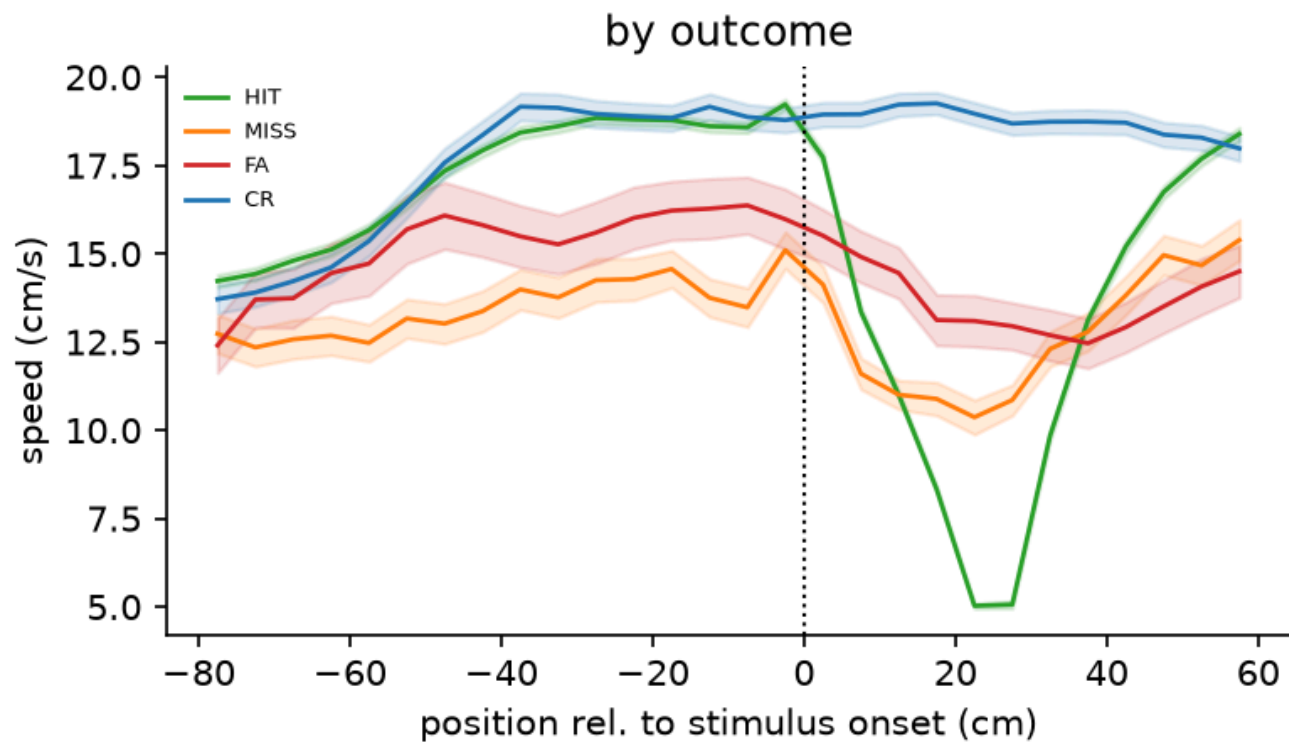
Psth time DP

Trial-aligned time -- DP (0 = stimulus onset)

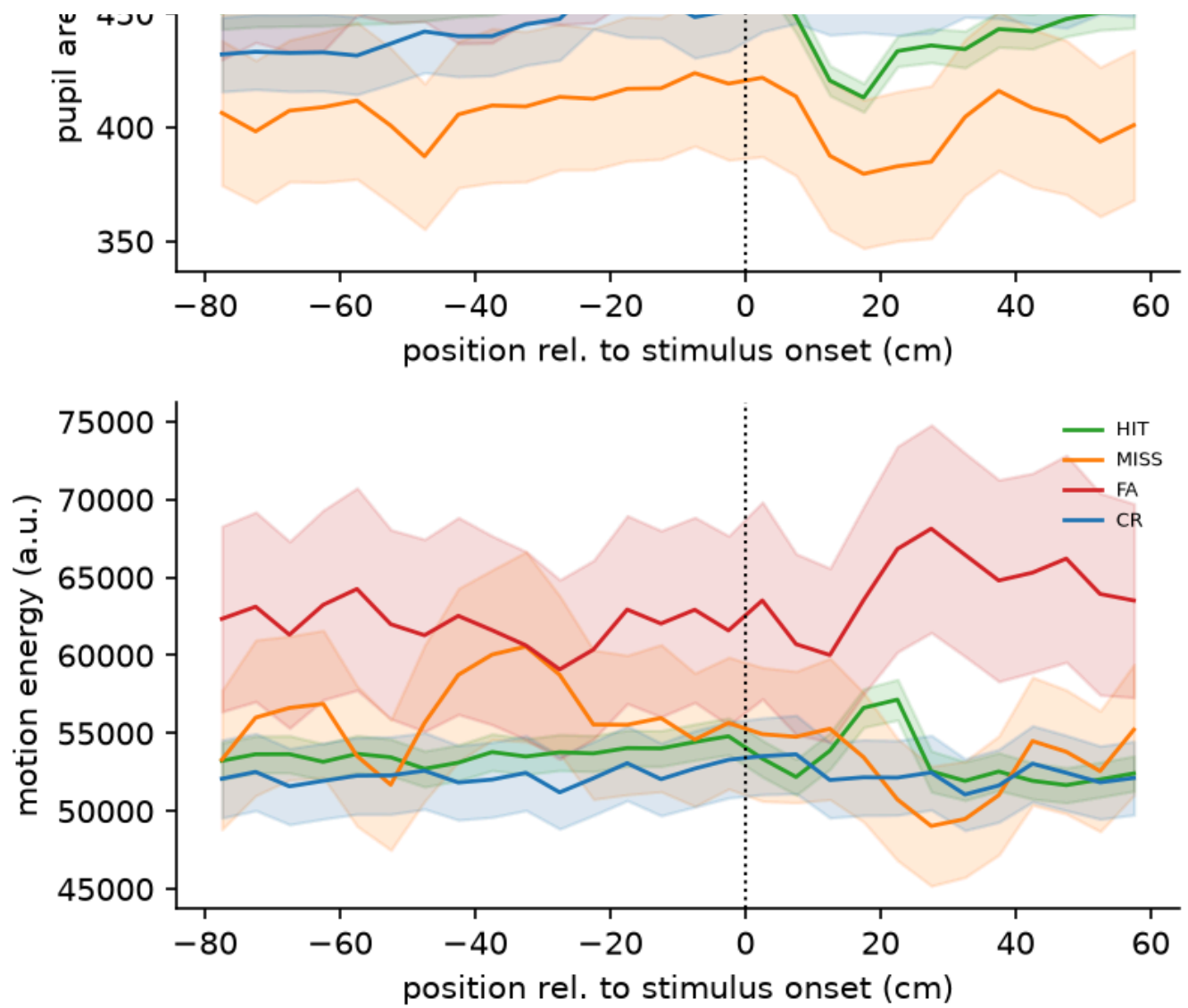


**Psth position DM**

Trial-aligned position -- DM (0 = stimulus onset)



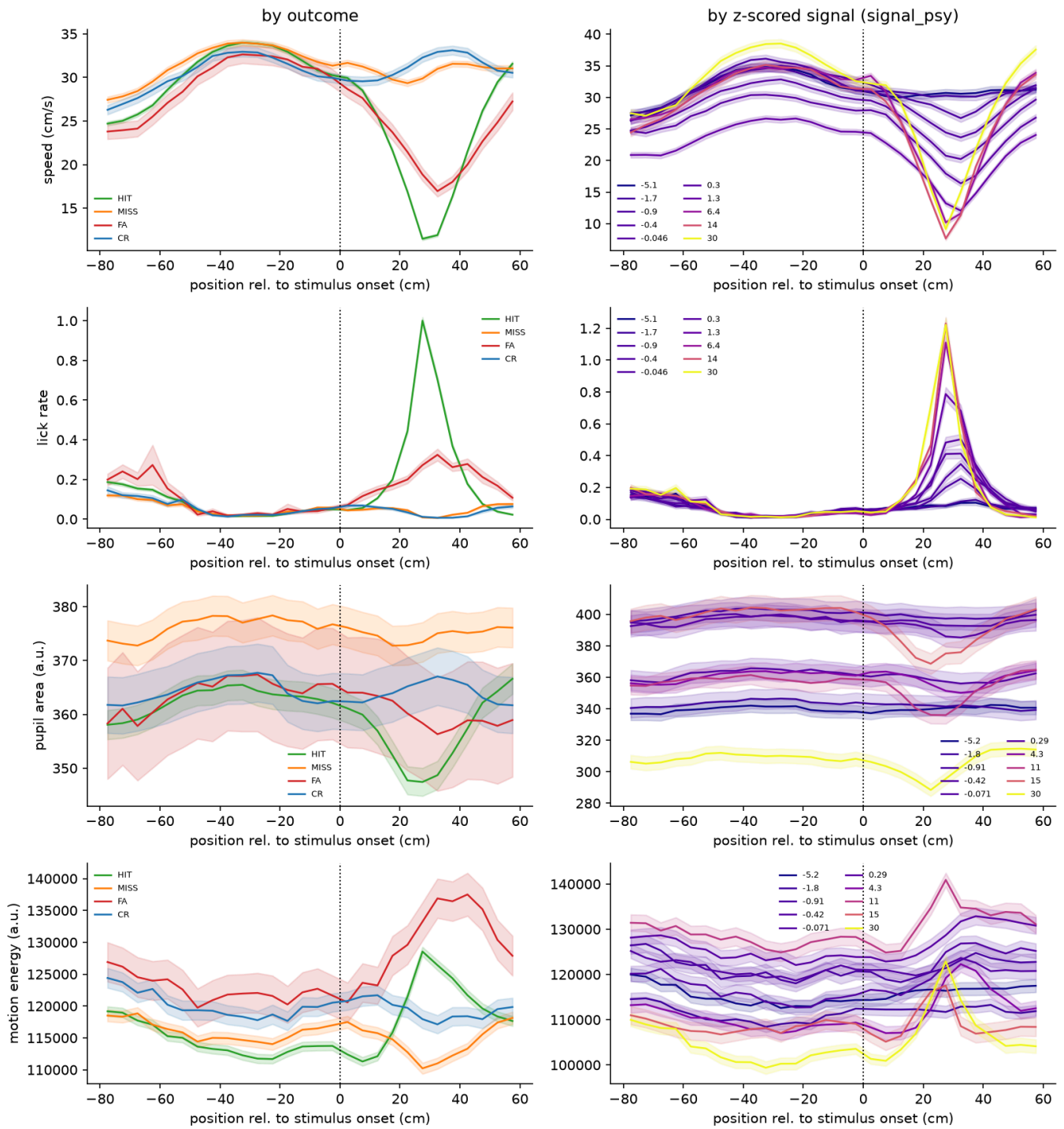




**Psth position DN**

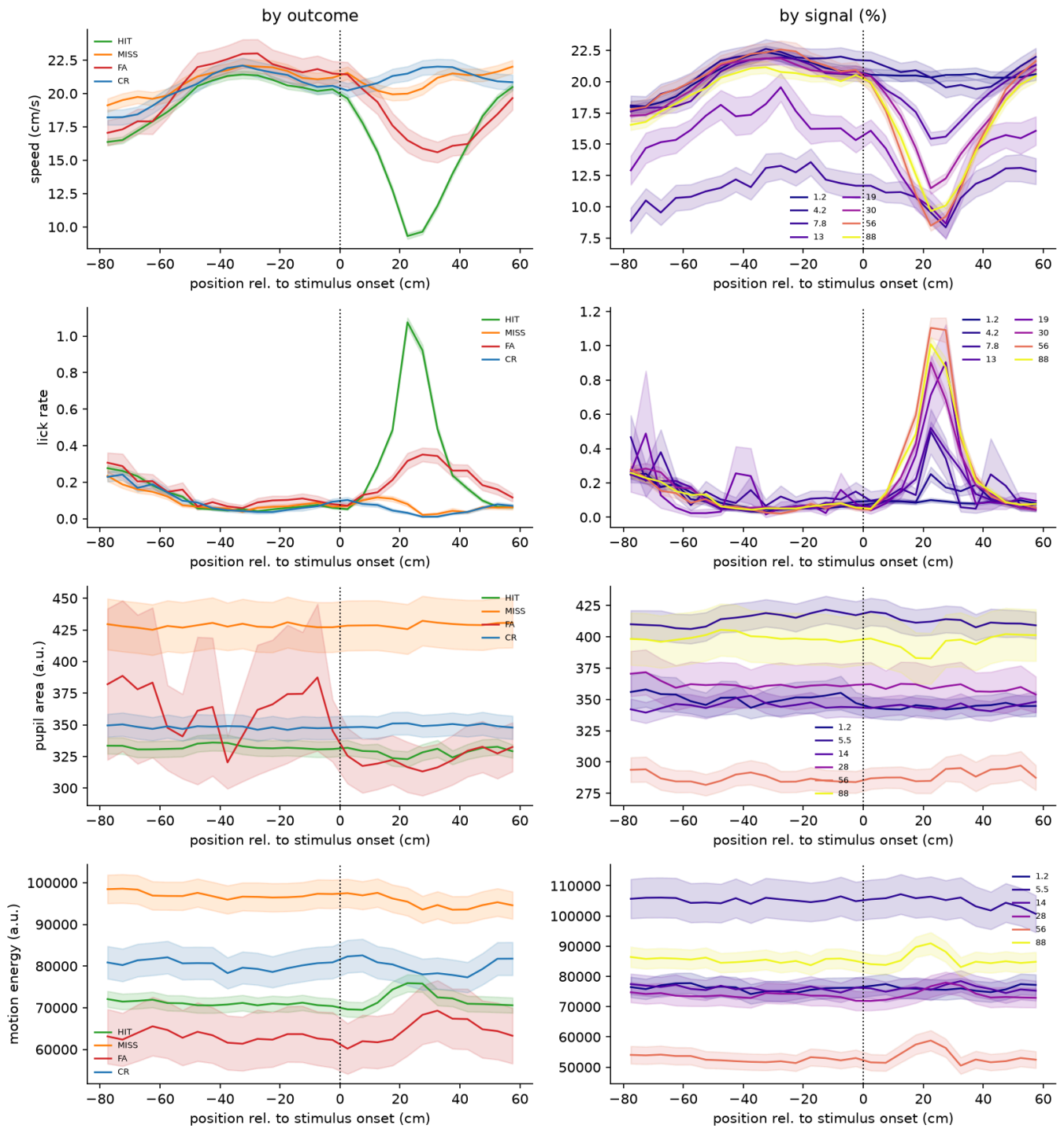


# Trial-aligned position -- DN (0 = stimulus onset)



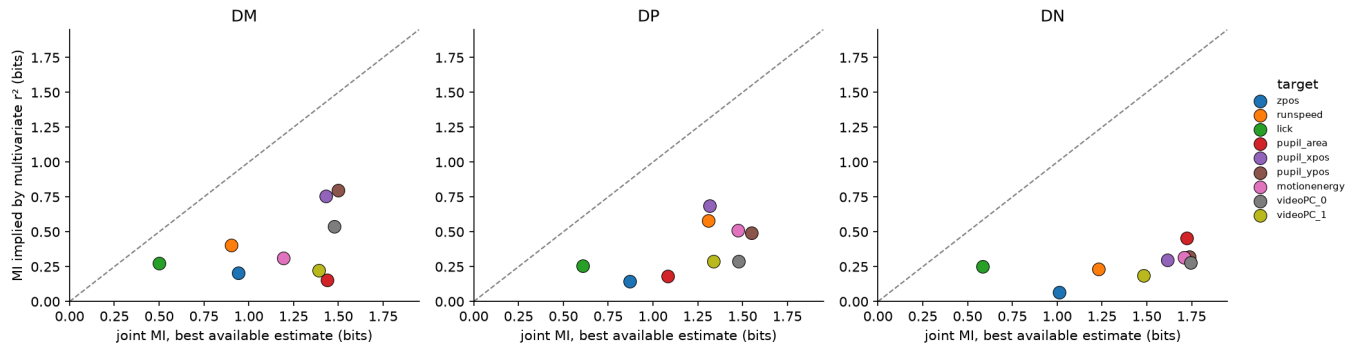
**Psth position DP**

Trial-aligned position -- DP (0 = stimulus onset)



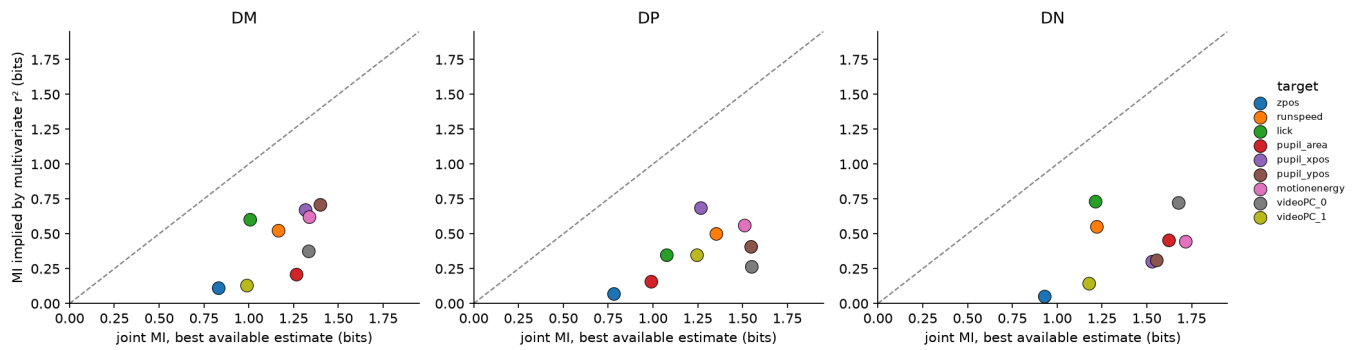
## Multivariate mi vs linear summary pre stim

Multivariate: does a linear model of ALL other variables capture the full relationship? -- pre\_stim window  
(color = target variable, predicted from every other variable jointly; dashed = linear matches MI exactly)



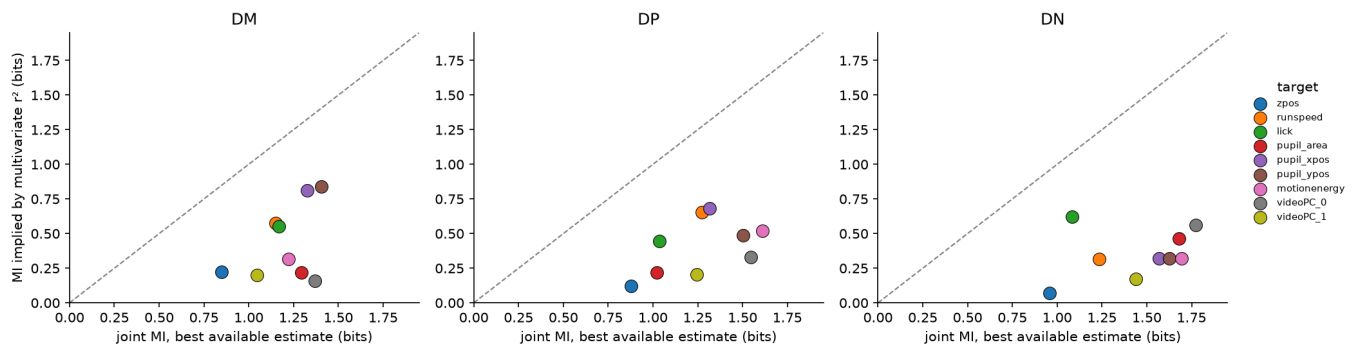
## Multivariate mi vs linear summary reward

Multivariate: does a linear model of ALL other variables capture the full relationship? -- reward window  
(color = target variable, predicted from every other variable jointly; dashed = linear matches MI exactly)



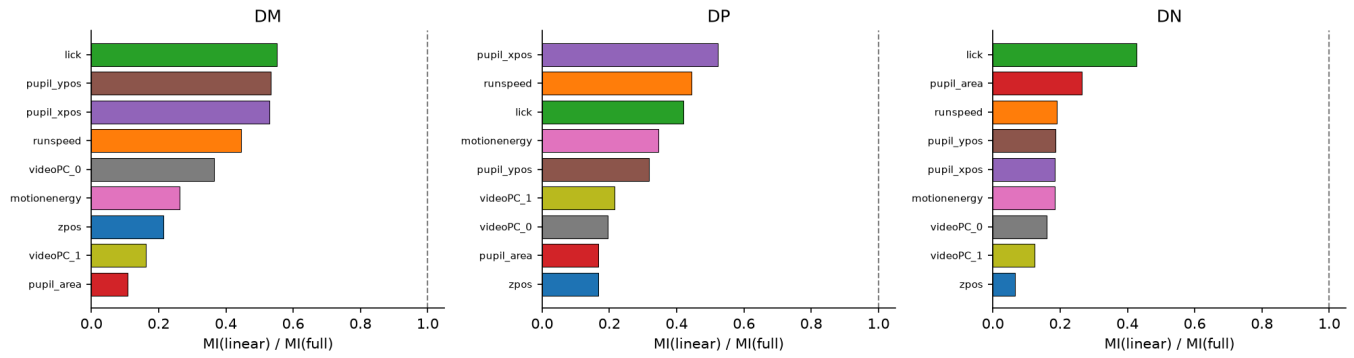
## Multivariate mi vs linear summary stim

Multivariate: does a linear model of ALL other variables capture the full relationship? -- stim window  
(color = target variable, predicted from every other variable jointly; dashed = linear matches MI exactly)



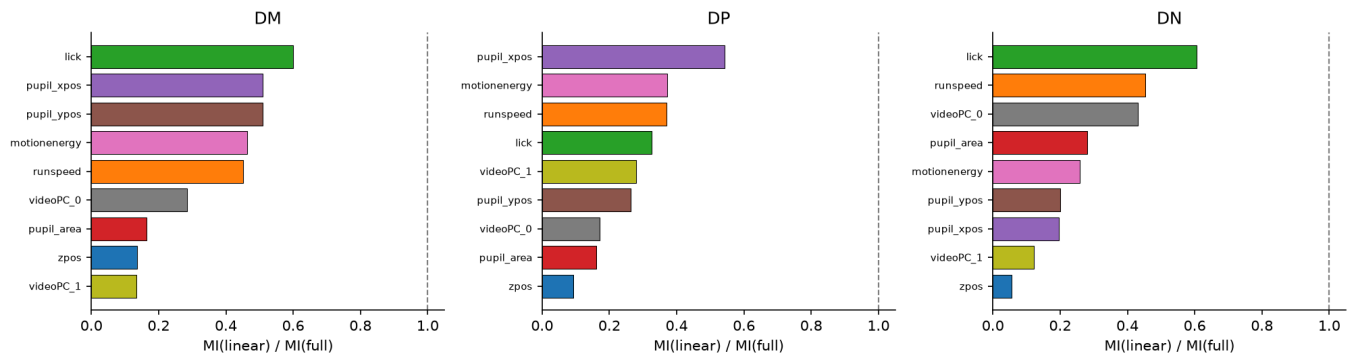
## 7a multivariate frac explained linearly pre stim

Multivariate: fraction of joint MI a linear fit captures, per target -- pre\_stim window



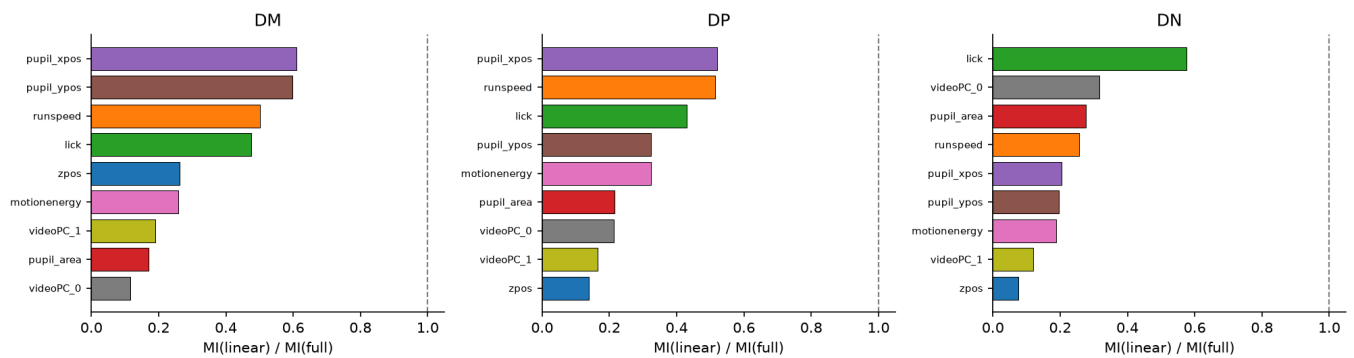
## 7a multivariate frac explained linearly reward

Multivariate: fraction of joint MI a linear fit captures, per target -- reward window



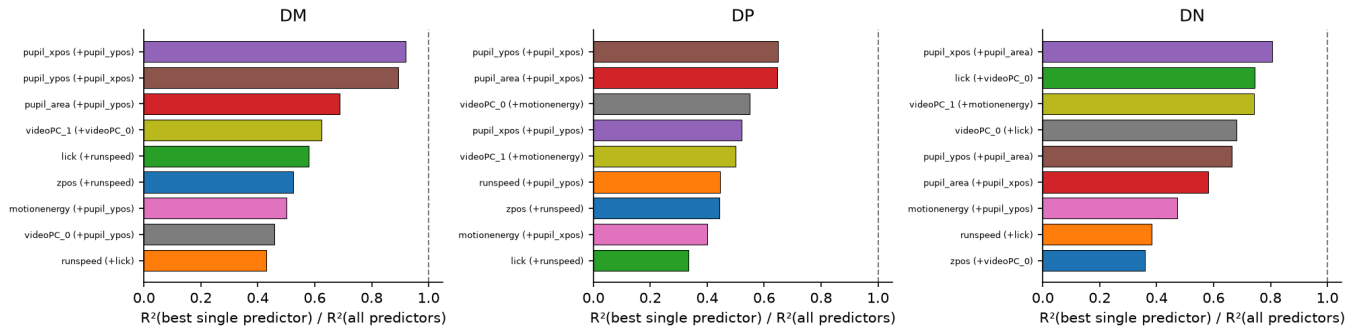
## 7a multivariate frac explained linearly stim

Multivariate: fraction of joint MI a linear fit captures, per target -- stim window



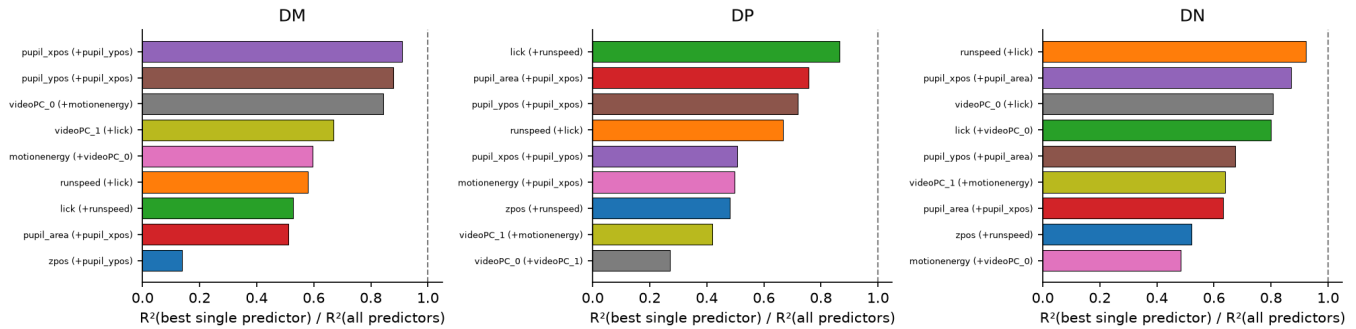
## Stepwise dominant predictor pre stim

Is the multivariate fit real synergy, or one dominant predictor? -- pre\_stim window  
(label = target (+its single best predictor); close to 1.0 = the rest add little beyond that one variable)



## Stepwise dominant predictor reward

Is the multivariate fit real synergy, or one dominant predictor? -- reward window  
(label = target (+its single best predictor); close to 1.0 = the rest add little beyond that one variable)



## Stepwise dominant predictor stim

Is the multivariate fit real synergy, or one dominant predictor? -- stim window  
(label = target (+its single best predictor); close to 1.0 = the rest add little beyond that one variable)

